

Q4 2024 Earnings Call

Company Participants

- Doug Bettinger, Executive Vice President and Chief Financial Officer
- Ram Ganesh, Head of Investor Relations
- Tim Archer, President and Chief Executive Officer

Other Participants

- Atif Malik, Citi
- Blayne Curtis, Jefferies
- C.J. Muse, Cantor Fitzgerald
- Harlan Sur, J.P. Morgan
- Joseph Moore, Morgan Stanley
- Joseph Quatrochi, Wells Fargo
- Krish Sankar, TD Cowen
- Melissa Weathers, Deutsche Bank
- Srinu Pajjuri, Raymond James
- Stacy Rasgon, Bernstein Research
- Timothy Arcuri, UBS
- Toshiya Hari, Goldman Sachs

Presentation

Operator

Good evening, and welcome to the Lam Research June Quarterly Earnings Call.

All participants will be in a listen-only mode. (Operator Instructions) After today's presentation, there will be an opportunity to ask questions. (Operator Instructions) Please note, this call is being recorded.

I would now like to turn the conference over to Mr. Ram Ganesh, VP, Investor Relations. Please go ahead.

Ram Ganesh {[BIO](#) [20521689](#) <GO>}

Thank you, and good afternoon, everyone. Welcome to the Lam Research quarterly earnings conference call.

With me today are Tim Archer, President and Chief Executive Officer; and Doug Bettinger, Executive Vice President and Chief Financial Officer.

During today's call, we will share our overview on the business environment, and we will review our financial results for the June 2024 quarter and our outlook for the September 2024 quarter. The press release detailing our financial results was distributed a little after 1:00 p.m. Pacific Time. The release can also be found on the Investor Relations section of the company's website, along with the presentation slides that accompany today's call.

Today's presentation and Q&A include forward-looking statements that are subject to risks and uncertainties reflected in the risk factors disclosed in our SEC public filings.

Please see accompanying slides in the presentation for additional information.

Today's discussion of our financial results will be presented on a non-GAAP financial basis unless otherwise specified. A detailed reconciliation between GAAP and non-GAAP results can be found in the accompanying slides in the presentation. This call is scheduled to last until 3:00 p.m. Pacific Time. A replay of this call will be made available later this afternoon on our website.

And, with that, I'll hand the call over to Tim.

Tim Archer {BIO 7502648 <GO>}

Thanks, Ram, and good afternoon, everyone. In the June quarter, Lam delivered another set of solid results, with revenues, profitability, and earnings per share all coming in above the midpoint of our guidance. Our CSBG business posted strong growth, with revenues up 22% sequentially, led by Reliant and spares.

On the manufacturing side, we achieved a key milestone in the quarter, with our Malaysia factory shipping its 5,000th chamber. This is the fastest ramp of a new manufacturing facility in Lam's history, and we remain on track to achieve our long-term cost-reduction goals through an expanded global manufacturing and supply chain footprint.

As previously communicated, 2024 is a year of strategic investment for Lam, where we are prioritizing product development for key technology inflections, global R&D infrastructure close to our customers, and digital transformation for operational efficiency at scale. We believe these investments will put Lam in a position to outperform as the industry moves into a period of multi-year WFE spending expansion.

Now, turning to WFE. We expect this year's spending to be in the mid-\$90 billion range. Our customer investment profile is generally unchanged from our prior view, apart from slightly stronger domestic China spending and additional demand related to the ramp of high-bandwidth memory, or HBM, capacity. We see foundry/logic, DRAM, and NAND investments all up on a year-on-year basis. Global spending on mature node technologies is expected to be roughly flat year-on-year. Looking ahead to 2025, we see a positive environment for continued growth in WFE spending.

The power of AI as a transformative business tool is still yet to be fully realized. Today, the focus on AI model training is driving strong demand for GPUs and HBM. However, as AI use cases expand, we believe Inferencing at the etch will spur content growth of low-power DRAM and NAND storage in enterprise PCs and smartphones. Investments for AI-enabled etch devices play particularly well to Lam's strengths. We anticipate that memory customers looking to scale capacity and lower bit costs will bias WFE spending toward technology upgrades of the installed base.

For NAND, the etch and deposition intensity of upgrades is significantly higher than in a greenfield investment. When you consider Lam's sizable installed base in memory, including roughly 7,500 high aspect ratio dielectric etch chambers for NAND alone, we are positioned to outgrow overall WFE when customers upgrade existing memory production lines to next-generation nodes.

Longer-term, etch and deposition are set to play an increasingly vital role in the industry's efforts to develop faster, more power-efficient and lower-cost semiconductors to serve AI-related applications. By delivering critical solutions for

atomic-level device scaling, new materials innovation, and advanced packaging integration, we see tremendous opportunity for Lam to expand our served market and increase our share at each successive process technology node.

To this end, our R&D focus is yielding exciting new products, including, this year, our first direct power coupled conductor etch tool with matchless power source and bias. Known as DIRECTDRIVE, this new power source uses solid-state drivers to stabilize the plasma in the etch chamber 500x faster than current industry standards. By combining direct power coupling with Lam's unique plasma pulsing capabilities, our latest conductor-etched systems are delivering best-in-class performance for newly emerging 4F-squared DRAM applications.

In 4F-squared devices, the nature of the bit line placement requires precise etching of ultra-small, high-aspect ratio silicon structures to avoid device shorts or leakage. With direct power coupling and plasma pulsing, Lam can etch the vertically oriented 4F-squared transistor architectures with unprecedented depth uniformity and profile control.

Similarly, conductor etch is becoming a critical enabler for EUV patterning for gate-all-around and DRAM, due to the need to reduce etch placement error. For nodes below 2 nanometers, the requirement is for roughly 40% tighter control than at 5 nanometers.

Our new conductor-etch tool delivers a 30% reduction in feature roughness, which is one of the main contributors to etch placement error. In addition, we can achieve one to two orders of magnitude improvement in defectivity for a given EUV dose, further helping customers reduce the overall cost and improve the capability of the EUV patterning process.

Turning to NAND. AI applications are driving demand for faster, higher-capacity enterprise SSDs. NAND makers are pursuing both vertical and lateral scaling of NAND arrays, as well as increasing bits stored per cell through implementation of QLC and TLC technologies. In support of these efforts, Lam is developing new dielectric etch and deposition capabilities.

Earlier today, we announced Lam Cryo 3.0, Lam's third generation of cryogenic-etch technology. Building on our learning from nearly 1,000 cryogenic-etch chambers running in NAND fabs worldwide, this new patented cryogenic-etch process delivers industry-leading control of the NAND memory channel hole profile.

When Lam -- Cryo 3.0 is deployed on our Vantex system, the etch are delivering the industry's highest available ion energy, we can create a 10-micron-deep channel hole that has a top-to-bottom profile deviation of less than 10 nanometers, or less than 0.1% relative to its depth. Such tight profile control allows customers to increase bit density by packing more cells per layer, while also having the flexibility to add more layers per tier.

Lam Cryo 3.0 also addresses our industry's need for more sustainable solutions, delivering a 40% reduction in energy consumption per wafer, and a 90% reduction in greenhouse gas emissions per wafer compared to non-cryogenic etchers.

Deposition technology is also advancing quickly to support increased bit density and lower cost through multi-tier stacking. Polysilicon and tungsten gapfill materials have typically been used to enable tier stacking in high-layer count NAND. Integration of these

materials, however, has resulted in poor control of critical dimensions and overlay, negatively impacting yield and performance.

Lam's innovative PECVD-based pure carbon gapfill process provides an attractive alternative material. With a unique combination of high etch selectivity, superior mechanical properties, and simplified dry post process removability, it also reduces the number of process steps required in some cases by approximately 50% compared to traditional approaches.

Overall, etch and deposition are becoming increasingly critical to addressing the complex semiconductor requirements of a growing AI environment. We are excited by the breadth of opportunities we see ahead for the company, especially those created by technology inflections to gate-all-around, backside power delivery, advanced packaging, and dry EUV resist processing. All of these are etch- and deposition-intensive, and each represents a \$1 billion or higher growth opportunity for Lam.

We look forward to sharing our progress on these fronts as well as our long-term financial model at our next Investor Day, which we are planning to uphold in February 2025.

With that, I'll turn it over to Doug.

Doug Bettinger {[BIO 16159085 <GO>](#)}

Great. Thank you, Tim. Good afternoon, everyone, and thank you for joining our call today.

We executed well on the June 2024 quarter. Our June quarter results came in above the midpoint or exceeded our guidance ranges for all financial metrics. We were pleased with the company's strong execution. For fiscal year 2024, we achieved the highest gross margin percentage since the merging of Lam with Novellus in 2013, coming in at 48.2%, and we generated quite strong free cash flow of approximately \$4.3 billion for 29% of revenue.

Let's look at the details of our June quarter results. Revenue came in at \$3.87 billion, which was an increase from the prior quarter and over the midpoint of guidance. Our deferred revenue balance at the end of the quarter five -- excuse me -- was \$1.55 billion, which is a decrease of \$194 million from the March quarter, related to revenue recognized that was tied to customer advanced payments. As we sit here today, I believe deferred revenue will remain stable at these levels for the foreseeable future.

Let's turn to the revenue segment details. June quarter systems revenue in memory was 36%, which was a decrease from the prior-quarter level of 44%. The decline in the memory segment was mainly attributable to DRAM. DRAM came in at 19% of systems revenue compared with 23% in the March quarter, as investments in mature nodes declined in the June quarter. DRAM revenue reached a new record in fiscal year 2024, with spending focused on DDR5 and HBM enablement, as well as on the 1Y node.

Nonvolatile memory came in at 17% of our systems revenue, which was down from the March quarter level of 21%. And just a reminder, we are characterizing one customer's investment in specialty DRAM as a nonvolatile investment, since it has a nonvolatile component in the device.

NAND revenue was at a low point for this year, and I expect NAND investment to gradually improve as utilization rates return to more normal levels and our customers slowly increased spending in conversions to 2xx and 3xx layer devices into the next year.

The foundry segment represented 43% of our systems revenue, which was roughly flat, with the percentage concentration in the March quarter of 44%. Growth in shipments for gate-all-around nodes was offset by a decline in mature node spending. The logic and other segment were 21% of systems revenue in the June quarter, up from the prior level of 12%. The increase was driven by strength in mature node spending in China.

With respect to the regional composition of our total revenue, the China region came in at 39%, down slightly from the prior-quarter level of 42%, and a little bit higher than our expectation from the previous earnings call. This was driven by domestic China spending. The next largest geographic concentration was Korea at 18% of revenue in the June quarter versus 24% in the March quarter. Taiwan was 15% of revenue in the June quarter, which was an increase from 9% in the March quarter.

The customer support business group revenue in the June quarter totaled approximately \$1.7 billion, an increase of 22% from the prior-quarter level and 14% higher than the June quarter in calendar 2023. CSBG revenue represented 44% of our June quarter revenues and reached the highest point since the end of calendar 2022, driven primarily by an increase in Reliant systems, followed by growth in spares.

Our Reliant systems revenue benefited from strength in domestic China spending for specialty and mature nodes. Spares revenue increased largely due to continued improvement in utilization at our memory customers as well as a little bit of inventory stocking. I do not think CSBG will grow modestly in calendar year 2024.

Let's look at profitability. Our June quarter gross margin came in at 48.5% at the top end of our guided range, and slightly down from 48.7% in the March quarter. June quarter gross margin benefited from continued improvement in factory efficiencies, which largely offset the headwind we saw in customer mix that we talked about on the last earnings call.

Operating expenses for the June quarter were \$689 million, down marginally from the prior-quarter amount of \$698 million. As Tim mentioned, we continued to prioritize spending in research and development to extend our technology differentiation as well as expand our product portfolio. I'd just point out that more than 70% of our total operating expenses were concentrated in research and development.

The June quarter operating margin was 30.7% above the guidance range, mainly because of that strong gross margin performance. Our non-GAAP tax rate for the quarter was 11.5%. We estimate the tax rate for the remainder of the calendar year 2024 to be in the low to mid-teens level, and this rate will fluctuate from quarter to quarter.

Other income and expense for the June quarter was approximately \$19 million in income, compared with \$10 million in income in the March quarter. The increase in OI&E was primarily the result of fluctuations in the fair value of our venture investments. And as we've talked about in the past, you will see variability in OI&E quarter-to-quarter.

Let's turn to the capital return. We allocated approximately \$382 million to share repurchases, and we paid \$261 million in dividends in the June quarter. During the quarter, we announced that our Board of Directors approved a \$10 billion share

repurchase authorization. We have \$10.8 billion remaining in the plan at the end of the June quarter. For fiscal year 2024, we returned \$3.7 billion, or 88% of free cash flow, which was in line with our long-term capital plans of returning 75% to 100% of free cash flow.

June quarter diluted earnings per share were \$8.14, close to the high end of our guidance range. The diluted share count was 131 million shares, on track with our expectations and down from the March quarter.

Let's turn to the balance sheet. Cash and cash equivalents totaled \$5.9 billion at the end of the June quarter, up a little bit from \$5.7 billion at the end of the March quarter. Days sales outstanding were 59 days in the June quarter, a slight increase from 57 days in the March quarter. June quarter inventory turns of 1.9x compared with 1.8x in the prior quarter. We are making progress in bringing inventory levels down, and we'll continue to work on this throughout the rest of calendar year 2024.

Our non-cash expenses for the June quarter included approximately \$79 million for equity compensation, \$74 million in depreciation, and \$14 million in amortization. Capital expenditures were \$101 million flat with the March quarter level, with spending mainly centered on lab investments in the United States and Asia, as well as manufacturing facilities in Asia, supporting our global strategy to be close to our customers' development and manufacturing locations.

We ended the June quarter with approximately 17,200 regular full-time employees, which was flat with the prior quarter.

Let's turn to our non-GAAP guidance for the September 2024 quarter. We're expecting revenue of \$4.05 billion, plus or minus \$300 million. Gross margin of 47%, plus or minus 1 percentage point. This gross margin decline is reflective primarily of an unfavorable quarter-to-quarter change in customer mix. I expect this change to continue to be a slight incremental headwind in the December quarter.

Operating margins of 29.5%, plus or minus 1 percentage point. Gross margin and operating margin including impact from ongoing transformation costs related to projects to improve our systems and operations. As we communicated at the beginning of the year, we're focused on re-engineering our business processes and systems to drive operational efficiencies and to implement AI at greater scale. And, finally, we're forecasting earnings per share of \$8, plus or minus \$0.75, based on a share count of approximately 131 million shares.

Let me wrap up. As we finished the first half of calendar year 2024, I was pleased that we were able to execute to the objectives we shared at the beginning of the year. We prioritized investment to extend our technology differentiation while driving operational improvements. We're encouraged that the spares business recovery is beginning and upgrade activity should improve as we exit the calendar year.

Longer-term, Lam is well positioned to capitalize the increase in etch and deposition intensity by delivering new capabilities of multiple new manufacturing inflections that we see ahead. We look forward to talking to you in February at our planned Investor Day about the long-term opportunities for Lam to continue our outperformance in the semiconductor industry.

Operator, that concludes our prepared remarks. Tim and I would now like to open up the call for questions.

Questions And Answers

Operator

(Question And Answer)

Thank you. We will now begin the question-and-answer session. (Operator Instructions) Your first question comes from Tim Arcuri with UBS. Please go ahead.

Q - Timothy Arcuri {BIO 3824613 <GO>}

Thanks a lot. Doug, I wanted to ask about the service system mix in the guidance. So you said that the service -- I thought you said that you now think it's going to grow modestly this year, but if I flatline service in September and December, it's up 8% year-over-year. So can you just clarify what you're thinking for service in the guidance mix?

A - Doug Bettinger {BIO 16159085 <GO>}

Yeah, Tim, we don't decompose the individual components of the guide. I was clarifying, we now expect CSBG to be up a little bit for the year. It was particularly strong in the June quarter. Whether it is up, down, or sideways from that as we go forward, I'm not going to give you individual components of the forecast, but I do think for the year, it's going to grow a little bit.

Q - Timothy Arcuri {BIO 3824613 <GO>}

Okay, great, Doug. Thanks. And then can you talk a little bit about just in DRAM, I think there's generally more excitement, Tim, about DRAM WFE than NAND, WFE among most investors out there about where it could go during this next peak. Obviously, you do very well in NAND, but in DRAM, you did talk about a lot of the investments which you're making.

Can you just talk about -- I know you're levered to the advanced packaging part of the HBM dollars being spent, but that's still a pretty small piece of it. So can you just maybe give a chance to kind of distill some of the view that you're not very levered to DRAM and give us a sense of maybe where you're investing and where you think you can gain share in DRAM? Thanks.

A - Tim Archer {BIO 7502648 <GO>}

Sure, thanks, Tim. And, as you said, we do very well in NAND, and we still think NAND's day is coming, as I said. We've seen some of those commentaries around enterprise SSDs, et cetera.

But on the DRAM side, the reason we highlighted the progress we're making, especially in this conductor etch, I mean, one, it's a new tool we've introduced. Its new capabilities that are very exciting for the industry and really targeted towards the types of ultra-small structures that are going to exist in future DRAM nodes going forward.

Lam is the global leader in conductor etch. And so we're applying all of that expertise and learning we have towards future DRAM challenges. And I think there's tremendous

opportunity for us on those applications, as I pointed out.

The other side of it, a lot of the excitement around DRAM is related to HBM. And there, as you commented, we play extremely well with our strong position in both TSV etching, as well as the TSV electroplating. And I think that we don't see any change in that strong position going forward.

So we get the benefit both from the scaling and architectural changes that are occurring in DRAM going forward, and from the advanced packaging and HBM-related expansion. And all of these, on both of those sides, are multiplied by the fact that you get fewer bits per wafer, and so everybody recognizes you're going to need a lot more DRAM wafers processed going forward. And, ultimately, that translates into more equipment from Lam.

Q - Timothy Arcuri {[BIO 3824613 <GO>](#)}

Thank you, Tim.

A - Doug Bettinger {[BIO 16159085 <GO>](#)}

Thanks, Tim.

A - Tim Archer {[BIO 7502648 <GO>](#)}

Thanks.

Operator

The next question comes from Krish Sankar with Cowen & Company. Please go ahead.

Q - Krish Sankar {[BIO 16151788 <GO>](#)}

Yes. Hi. Thanks for taking my question. My first one is for Doug. I think, Doug, you gave some color on China. Kind of curious how to think about China into the back half of this calendar year and into calendar '25. And along the same path, you kind of mentioned that December quarter, there could be a slight more gross margin headwind. Is there a way to quantify how much that -- how many basis points that headwind would be in December compared to the 47% in September? And then I had a follow-up with Tim.

A - Doug Bettinger {[BIO 16159085 <GO>](#)}

Yes. Sure. Krish, I'll just remind you what we said last quarter. It hasn't really changed from this quarter. And what that statement was that, for the year '23, '24 China's up. However, it is a somewhat first-half-weighted year this year, as opposed to last year, it was somewhat second-half-weighted. I'm not communicating, hey, it's going away. It's not going away.

It's just the spending, because sometimes these customers are a little bit bigger than a breadbox. It can be a little bit lumpy. And that's very much what we're seeing in China. I'm not ready to tell you exactly what next year looks like from the China region, but I do think it's going to be a pretty solid year. Again, it's not going away, and it's too soon for us to quantify things for next year, but '25 should be a pretty decent year in China, Krish.

Q - Krish Sankar {[BIO 16151788 <GO>](#)}

And then, Doug, any color on the December quarter gross margin?

A - Doug Bettinger {BIO 16159085 <GO>}

I'm not going to give you a number, Krish. But I've been signaling for a while that because of customer mix, margin will have a little bit of some headwind going into the second half of the year. I just guided you to 47% in September and suggested that there might be a little bit of incremental headwind into December because of customer mix, is what I said in the script.

Q - Krish Sankar {BIO 16151788 <GO>}

Got it. Got it. Thank you for that, Doug. And then, Tim, just a quick follow-up. When I look at all the upcoming tech inflections, like gate-all-around, backside power delivery, maybe down the road 3D DRAM. You spoke within that the transition to 4F-squared DRAM from 6F-squared, I'm kind of curious, is that really that material? And if so, is there a way to seize the opportunity for Lam at 4F-squared? You spoke a little bit about conductor etch. I was just kind of wondering if you could give some more color around how to quantify that number for the 4F-squared architecture transition. Thank you.

A - Tim Archer {BIO 7502648 <GO>}

Sure, well, I don't think we're prepared to quantify it for you today, but I think that, what, my comment was, though, we see each of these inflections, and 6F-squared to 4F-squared is a technology inflection that brings with it some important changes. I mean, one, the architectural layout of the device itself puts additional requirements on etch, which I think we're very well suited to serve, and that's why we've been developing new conductor etch capabilities to target those new requirements. So there is some incremental opportunity there.

Clearly, the jump to 3D NAND is a much bigger step-up in etch and dep intensity, but our goal is to increase our SAM and grow our share at every technology node, so we look at whatever is the new requirement and how we can best address that. You also look, as you look at DRAM going forward, another thing that's happening, whether it's 6F-squared as you move to 4F-squared, is the implementation of more EUV layers and how Lam plays in EUV.

Again, anything where effectively pattern transfer etches, the feature sizes are getting smaller, precision is required, these are the kinds of high-tech etches that Lam excels at, and so we look at participating in those.

And then on the deposition side, of course, we've talked about things like our dry EUV resist process and how that plays into EUV as DRAM and foundry/logic transition from EUV to high-end EUV. And so we're just looking at every technology node as an opportunity for us to gain.

Q - Krish Sankar {BIO 16151788 <GO>}

Thank you.

A - Doug Bettinger {BIO 16159085 <GO>}

Thanks, Krish.

Operator

Your next question comes from Srini Pajjuri with Raymond James. Please go ahead.

Q - Srini Pajjuri {BIO 21216060 <GO>}

Thank you. Tim, I have a question on DRAM. Obviously the recovery has been ongoing, and HBM is a secular driver that you talked about, and you do have a very strong position in that market as well. Then I look at your revenue. I think it peaked around December of '23, and it's been kind of declining on a sequential basis. I'm guessing some of that is maybe mature node DRAM.

Just wondering if you're kind of at the bottom, and then given all the talk about HBM spending, I would think that it's going to kind of, it should at some point come back strongly. So I just want to hear your thoughts on why it's been declining and how should you think about in particular in DRAM revenue.

A - Tim Archer {BIO 7502648 <GO>}

Yeah, I think you generally have it pretty correct. We had talked about the fact that some mature node DRAM spending was a little bit heavily concentrated in the second half of last year through early part of this year even. And as that came off, there was some of a reset in what we would call kind of traditional or the conventional DRAM. That is being picked up at some rate by the growth in HBM. But HBM itself is still in its ramping phase.

And I think that as we look into 2025, it becomes an even bigger driver of wafers in DRAM. And so I think that explains probably the profile. And I think that as we look forward, the secular driver of HBM, the impact on wafer, effectively how many wafers it requires to produce that number of bits due to the die size and due to the complexity of stacking these DRAMs means that we see DRAM demand for DRAM equipment continuing to grow through 2025 and probably well beyond that.

Q - Srini Pajjuri {BIO 21216060 <GO>}

Okay. Got it. Thank you. And then on the CSBG business being up 20% sequentially, 22%. I know you don't want to give us guidance going forward, at least for next quarter. I'm just curious about the sustainability of some of the trends that you're seeing, Tim. And then what does that mean for the overall WFE? Is this a prelude to something? And is this just the utilization improving? And then, does this usually follow in terms of WFE increasing and in terms of new tech migrations or capacity additions? So any color on that would be helpful.

A - Tim Archer {BIO 7502648 <GO>}

Sure. It's a good question. And I guess, again, reminding people that the CSBG business includes our Reliant business, which sells into mature nodes, it includes spares, it includes upgrades and services. And so each of those components move somewhat differently. And we talked about this quarter particularly being strong as a result of Reliant and spares.

We are starting to see a pickup in utilization in the memory fabs, as we've talked about. And I've talked a little bit about the fact that, as we look forward, we think that upgrades will begin to become a much more prominent part of our customers' WFE spending if they look to upgrade memory fabs that really haven't been upgraded in quite some time

because of the severe downturn that we've seen in those markets over the last few years.

And so I do think, going forward, you see a little bit more of a balance between those different segments, upgrades coming up stronger, and spares continuing to grow simply because our installed base itself continues to get bigger. Traditionally, we would have always said that we would expect the CSBG business to grow every single year, and that's simply effective every year we ship more tools, and those tools then require services and spares and basically present new opportunities for Lam to capture revenue from those systems.

So I think long-term, CSBG will returning back to that growth. And next year probably much more biased towards the upgrades business as customers start to do brand new fab upgrades.

Q - Srini Pajjuri {[BIO 21216060 <GO>](#)}

Got it.

A - Doug Bettinger {[BIO 16159085 <GO>](#)}

Thanks, Srini.

Q - Srini Pajjuri {[BIO 21216060 <GO>](#)}

Thanks, Tim.

Operator

Your next question comes from C.J. Muse with Cantor Fitzgerald. Please go ahead.

Q - C.J. Muse {[BIO 6507553 <GO>](#)}

Yeah, good afternoon. Thank you for taking the question. I guess first question was hoping to focus on gross margins. A couple quarters ago, Doug, you talked about kind of looking back to the June kind of '23 quarter as normalized, but given the guide today, it sounds like that was conservative and it's a higher number.

So just as you think about calendar '25, as you get to kind of a normalized China mix and you normalize to Reliant, what would be kind of the base level we should be thinking about for gross margins? And then can you talk to what kind of accretion we should be thinking about related to Malaysia and/or some of these higher-margin upgrade drivers?

A - Doug Bettinger {[BIO 16159085 <GO>](#)}

Yeah, C.J., I mean, you've alluded to some of the things that move gross margin, obviously. Yeah, I had previously anchored you and others with respect to that June quarter before the China mix improved or strengthened, I guess, maybe not improved.

As the baseline in I guided it down a little bit in September, described customer mix softening a little bit relative to moving that, and I'm not suggesting, hey, a little bit more in December, potentially. It's all about customer mix. Too soon for me to guide you for next year, but the things you should be thinking about is what does that customer mix look like next year?

I'm not sure yet, and I'm not ready to point you to numbers, but what will begin to show up in a more significant fashion is the accretion from those Asia factories as we ramp output. That will be a benefit to gross margins. So those are the moving pieces to be thinking about. The customer mix, I'm not entirely sure, but as we see a likely WFE environment next year that's somewhat stronger, increasingly the incremental volume will be supported from those Asia factories, which should be beneficial to gross margins, C.J.

Q - C.J. Muse {BIO 6507553 <GO>}

Very helpful, and I guess as my follow-up in your prepared remarks, you spoke to the 7,500 high aspect ratio etch chambers installed in the NAND industry, and just curious, as you see upgrades there, what kind of growth could that add to overall NAND WFE, specifically to you guys? Is there kind of a percentage we should think about? Any help there would be great.

A - Tim Archer {BIO 7502648 <GO>}

We haven't quantified that, but the reason I included it was simply to point out the installed base itself becomes a powerful driver of revenue during those upgrade cycles.

And we do think that the next phase, we've seen higher memory fab utilization, particularly we talked about NAND beginning to improve last quarter, and that's seems to be continuing as we move through the remainder of this year, get to next year and the upgrades start in earnest, those tools are -- those tools represent opportunities for Lam to help our customers achieve it's both a technology upgrade and a bit cost reduction as they move forward and accrues quite a lot of revenue for Lam relative to the amount of WFE spend.

I'll remind people the WFE Lam's capture rate of spending in an upgrade is significantly higher because etch and deposition represents so much of the upgrade. So that was the reason we pointed out the size of our installed base.

Q - C.J. Muse {BIO 6507553 <GO>}

Thanks so much.

A - Doug Bettinger {BIO 16159085 <GO>}

Thanks, C.J.

Operator

Next question comes from Stacy Rasgon with Bernstein Research. Please go ahead.

Q - Stacy Rasgon {BIO 16423886 <GO>}

Hi, guys. Thanks for taking my question First, I wanted to ask, like, if the NAND business next year is primarily driven by upgrades, what does that imply for growth? Like, would it be conceivable that NAND WFE could double year-over-year in calendar '25? It was purely upgrade-driven, or would you need capacity additions to get there, and are you seeing any signs at all of capacity additions right now? Doesn't sound like it.

A - Tim Archer {BIO 7502648 <GO>}

Well, Stacy, what I would say is that obviously we're not going to guide what NAND WFE is next year. Frankly, I think it's still a developing story, but what we're trying to say is that, as customers move to upgrades, whatever WFE is spent, Lam will be the primary beneficiary of that WFE spend.

And so that's a year in which my comment was we would be confident that we would outgrow WFE in the NAND space if it was primarily upgrade spend and upgrades represent a tremendous efficient way for customers to essentially advance their technology and lower their costs.

And so we do think that will be the next phase of NAND investment based on our thoughts.

Q - Stacy Rasgon {BIO 16423886 <GO>}

Got it. Got it. I mean maybe to follow up on that just a little bit. I mean if I -- if you look at your current like NAND outlook for this year, would you say that that outlook has gotten better or worse or stayed the same versus, like, 90 days ago?

A - Doug Bettinger {BIO 16159085 <GO>}

Probably hasn't changed much, Stacy. This is Doug. Maybe a little bit better. We're trying to see a little bit of an uptick in utilization, but I don't think it's meaningfully different, Stacy.

Q - Stacy Rasgon {BIO 16423886 <GO>}

So I got it's a kind of in the noise?

A - Doug Bettinger {BIO 16159085 <GO>}

Yeah, kind of in the noise.

Q - Stacy Rasgon {BIO 16423886 <GO>}

Okay. That's helpful. Thank you, guys. Appreciate it.

A - Tim Archer {BIO 7502648 <GO>}

Thanks, Stacy.

A - Doug Bettinger {BIO 16159085 <GO>}

Thanks, Stacy.

Operator

The next question comes from Harlan Sur with J.P. Morgan. Please go ahead.

Q - Harlan Sur {BIO 6539622 <GO>}

Yeah, good afternoon, thanks for taking my questions. Given the strength in CSBG, it looks like utilization by your customer base continues to rise. Did that also broaden out to start to include not just leading-edge logic/foundry, DRAM, and NAND, but maybe also start to include mature and specialty fabs as well? Or are at a minimum mature and

specialty utilizations at least stabilizing in line with some of the cyclical improvements that we're seeing in the semi industry?

A - Doug Bettinger {[BIO 16159085](#) <GO>}

Harlan, the mature node stuff is still pretty soft, frankly. And I think you understand what's going on. If you just listen to everybody else's earnings calls in the analog, industrial, automotive space, there's still a lot of inventory out there. It's still relatively soft. The statements we're making around utilization have more to do with what we're seeing in the memory fabs, quite frankly.

Q - Harlan Sur {[BIO 6539622](#) <GO>}

I appreciate that. And then on HBM and advanced packaging, I mean, last night, we heard AMD talk about supply dynamics being tight on their AI GPU supply next year, CoWoS, and HBM. On the custom ASIC front, we hear companies like Broadcom keep getting upside orders from their AI customers like Google.

Last quarter, you talked about doing a \$1 billion in advanced packaging and HBM revenues this year. Has that number moved higher, and is the team capacity-constrained on advanced packaging systems? And are your lead times here for those tools starting to stretch out?

A - Tim Archer {[BIO 7502648](#) <GO>}

Well, it has moved higher, and we're not going to re-quantify it just yet. But it is -- we're seeing very strong demand in those areas. I talked about the expansion of our global manufacturing supply chain footprint, and obviously that's giving us more flexibility than we had during the last ramp.

Our goal through all those investments was to be able to respond in this next few years of expansion better than we did in the expansion that we saw right around the time of COVID. And so I think that that will position us.

Sure, you're always a little bit short, and customers always drop in tools within your lead time, which keeps you busy. But I think that we're doing quite a nice job responding to the urgent requests from our customers. We actually like this. I would say, generally from running the business, we like this environment where all parts of our business are a little bit supply-constrained.

I mean, you hear a lot of our customers talking about being cautious about adding capacity, all our customers talking about having a little trouble getting tools. I think that's a good place for us to be because it means that I think we're setting up for a more manageable long-term ramp of demand than sort of a short spike followed by, again, periods of digestion that always create a little bit of chaos in the industry.

Q - Harlan Sur {[BIO 6539622](#) <GO>}

Yes. Thank you, Tim. Thanks, Doug.

A - Doug Bettinger {[BIO 16159085](#) <GO>}

Thanks, Harlan.

Operator

Your next question comes from Atif Malik with Citi. Please go ahead.

Q - Atif Malik {BIO 7312618 <GO>}

Hi. Thank you for taking my question. Doug, if I look at the 2023 year-over-year China sales growth among the big five equipment makers, all of them are up quite well. Many of them are up like 250%, and the U.S. peers are up in teens or 20%. But you guys were down 11% total China sales in 2023, and this year you're expecting China sales to be up.

So I'm just trying to understand the dynamics last year. Were this just a function of maybe NAND spending and the NAND project not being active, or are there competitive elements in China that are working against you?

A - Doug Bettinger {BIO 16159085 <GO>}

Atif, I'll remind you that perhaps our largest customer got restricted when the regulations came out, our NAND customer in China. That customer was pretty strong in 2022, went away in 2023. So the year-over-year comparisons you're making, you've got to factor that in. And then the strength we're seeing in 2023-2024 is a different mix entirely. Really not any NAND in China to speak of, at least not domestic China. I don't know if that helps you, but make sure you're thinking about that.

Q - Atif Malik {BIO 7312618 <GO>}

Yeah, it all makes sense. That's helpful. And then on the cryo improvement, Tim, that you mentioned, are those improvements or process tool of record is going to solidify your market share next year or the year after?

A - Tim Archer {BIO 7502648 <GO>}

Well, I think that all of these things, I mean, when you introduce something new, I mean, I think what people kind of lose sight of is, generally we're working several years ahead with our customers on R&D. I talked about the investments we're making where we're building labs close to our customers in different geographies, that's because in many cases we're engaging those customers a good five years ahead of production implementation. That's not to say Lam Cryo 3.0 is going to take five years to get into production, but it's not a technology that's ramping, say, this year, but it really is looking out at the needs of our customers one to two generations out and really solving their difficult etch challenges.

So, sometimes when the benefits are so good, and I talked about the fact that we get about 2.5x the etch rate, tremendous profile control, customers will often pull that in sooner, but really this is designed for kind of the 400-plus layer, may ultimately end up being pulled in earlier than that, but that's where you really start to see the needs for this kind of capability.

Q - Atif Malik {BIO 7312618 <GO>}

Thanks. That's exactly what I was asking about. If it's 400 layers or 200 layers, but sounds like 400 layers.

A - Tim Archer {BIO 7502648 <GO>}

Yes, and I think that what we, you'll see in our press release today, we talked a little bit about we're trying to chart the path across not only etch but also our deposition films towards where the industry needs to go to get to 1,000 layers, because we truly see over the next decade that that's where you want to get in terms of satisfying bit density and cost as NAND demand continues to expand with AI.

Q - Atif Malik {BIO 7312618 <GO>}

Great. Thank you.

A - Doug Bettinger {BIO 16159085 <GO>}

Thank you.

Operator

Next question comes from Toshiya Hari with Goldman Sachs. Please go ahead.

Q - Toshiya Hari {BIO 6770302 <GO>}

Hi, guys. Thank you so much for taking the question. I joined late, so I do apologize if these questions have been addressed. Just on the third generation Cryo tool, Tim, that you spoke about, how is this technology or tool fundamentally different or better than your nearest competitor? I know you just spoke to some of the characteristics, but if you can clarify that for us to the extent you're comfortable, that would be really, really helpful.

And then my second question, again, on the CSBG side, probably one for Doug, and again, you may have addressed this. For the full year, calendar year, I think you previously said flattish plus or minus. Is that still the view, or given the strength you saw in June, is that - should we be thinking about a higher growth rate for the full year? Thank you.,

A - Doug Bettinger {BIO 16159085 <GO>}

Thanks, Toshiya. Maybe I'll take that one first. And then I'll let Tim comment on the 3.0 those stuff. Yes, you might have missed my scripted statements. As we sit here today, we now expect that for 2024 CSBG, the word I used was modestly grow, this year versus last year.

And part of that is, we saw particular strength in the June quarter in Reliant. A little bit of improvement in spares, and as we think about the utilization trends that are likely occurring with our memory customers, I think spares continues to be decent, and we're optimistic that we'll start to see some of the upgrades, Tim, that we've been talking about for a while.

I'll hand it over to Tim on the 3.0 stuff.

A - Tim Archer {BIO 7502648 <GO>}

Sure. Thank you. I think, obviously, what I would just start with is the biggest difference between what we're delivering with Lam Cryo 3.0 and what our competitors do is in the results on the wafer, which we talked about pretty amazing 10-micron-deep holes with less than a 10-nanometer taper from the top to bottom at etch rates that are 2.5x conventional etching.

So it's the results that are pretty amazing. We talked about the fact that this is based on some new surface chemistries that are enabled in our tool, and there's a whole combination of hardware issues -- hardware configurations and capabilities in our tool that I think allow us to achieve that result.

And I can't go into all of those details today, but it is -- I did allude to one of them, which is on our Vantex system, the chamber design allows us to deliver a significantly higher ion energy than what is available from any other system available in the semiconductor industry. And that does play some role in etching these very, very deep holes with near-perfect verticality. So that's about all I can say today.

Q - Toshiya Hari {BIO 6770302 <GO>}

Thank you. Appreciate that.

A - Doug Bettinger {BIO 16159085 <GO>}

Thanks, Toshiya.

Operator

Thank you. The next question comes from Joe Moore with Morgan Stanley. Please go ahead.

Q - Joseph Moore {BIO 17644779 <GO>}

Great. Thank you. I wanted to ask you, I mean, there's been a number of press concerns about export controls, was talking about the foreign direct product rule, which doesn't seem like it would affect you, but also talking about entity list. And I'm just wondering -- obviously we don't know what would happen with any of that, but are you seeing any different behavior from your China customers? Are you seeing them push things in or pull things or pull things -- push things out because of any of those anxieties?

A - Tim Archer {BIO 7502648 <GO>}

Yeah, Joe, I think that obviously we don't know exactly what's going to happen either, just as you said, and so we can't really speculate on that. I have mentioned in the last couple of calls that there are ongoing discussions all the time with the U.S. government and regulatory agencies. We're part of those discussions and will continue to be.

I think in terms of change in behavior by any of our customers, I don't think it's something that's noticeable, nor would it be something that we would be able to easily react to.

We've talked about how we deal with some of these new customers that emerge with downpayments and other things to make sure that we understand the -- those customers as viable customers, but beyond that, we service them like others at this point as long as we can ship to them. And I would say lead times and responsiveness from our perspective is the same as we treat any customers at those sites.

Q - Joseph Moore {BIO 17644779 <GO>}

Very helpful. Thank you.

A - Doug Bettinger {BIO 16159085 <GO>}

Thanks, Joe.

Operator

Next question comes from Blayne Curtis with Jefferies. Please go ahead.

Q - Blayne Curtis {BIO 15302785 <GO>}

Hey, thanks. Let me ask a question. I know you had a couple on this, Doug, on the China business. I'm just kind of curious. I think you qualified as a solid year next year, and I just didn't know what that meant. So I know you've been hesitant to kind of call China. I think you called it like flat, plus or minus, maybe up or down last time.

Do you feel better about it, I guess, outside of this June? That's the other kind of part of the question. Just you can qualify a little bit. I mean, you should have some idea of what you're going to ship to pay ahead. So is June kind of a cleanup, and it might be a little bit lumpy? And -- Or is China actually trending a bit better for you?

A - Doug Bettinger {BIO 16159085 <GO>}

I don't know, Blayne, that I'm trying to communicate anything any different than we said on the last call, to be honest with you. We described this year as somewhat first-half-weighted. Really no change to that.

The June quarter was maybe a little tiny bit stronger in China, but only a little tiny bit. It's too soon for me to -- for us to quantify 2025. But what I would tell you is I expect next year to be a solid year in terms of spending in China. I'm not going to give you a number yet, because I'm not completely sure. But what I wouldn't want anybody to think is it's going away, because it's not.

Q - Blayne Curtis {BIO 15302785 <GO>}

Got you. And then, I'm just kind of curious, the broad strokes was growth for next year. I know you don't want to give a forecast. But in terms of the moving pieces there, I mean, it's pretty clear, leading-edge is strong, DRAM spend is strong. I'm just kind of curious, as you look into that forecast, if you're willing to venture kind of a view on the NAND business.

A - Doug Bettinger {BIO 16159085 <GO>}

Yes, I think NAND spending next year has to be greater than it is this year, right? We're off two years of quite low spending in NAND. I don't know what magnitude. I expect next year -- We expect next year you're going to see a lot of upgrades in NAND, but too soon for me -- for us to give you a number, but I would be shocked if it's not stronger than it is this year. It has to be.

Q - Blayne Curtis {BIO 15302785 <GO>}

Thanks, Doug.

A - Doug Bettinger {BIO 16159085 <GO>}

Yes. Thanks, Blayne.

Operator

The next question comes from Joe Quatrochi with Wells Fargo. Please go ahead.

Q - Joseph Quatrochi {BIO 18961101 <GO>}

Yeah. Thanks for taking the question. I wanted to try it on the NAND side again. If we just think about your prior peak NAND revenue ex your customers that are obviously now restricted, can capacity upgrade to just higher layer counts support your return to those levels, just given your higher share of etch and the -- the higher etch and dep intensity for the transitions?

A - Doug Bettinger {BIO 16159085 <GO>}

Joe, if it's just an upgrade, your spending will be lower than when capacity gets added. Obviously, our share of wallet will be greater. The fact that we lost a pretty large NAND customer in the China region, hard to replace that. I'm not ready to tell you what kind of next year is relative to previous peaks, but I know next year is going to be a stronger year in NAND than it is this year, for sure.

A - Tim Archer {BIO 7502648 <GO>}

Yes. And, Joe, I think that, obviously, if we're talking next year, it's very specific to the demand environment and the upgrade business in 2025. Longer-term, I mean, we've laid out in our view is that NAND spending rises. And so, it's just a matter of the time frame you're looking at.

And, from the standpoint of Lam's business, the etch and dep components and the complexity of tier stacking, the precision that's required for implementation of QLC and TLC technologies, all of these skew towards Lam's technical strengths and also SAM expansion opportunities. And I talked about the PECVD pure carbon gapfill process, which is a new addition to the portfolio for NAND scaling technology going forward.

And so, if I go back and I look at where we were in the portfolio we had to sell, when people used to think of NAND as a very, very strong business for Lam, we've expanded that portfolio quite substantially with the gapfill, the backside stress management, the ALD oxide gapfill process, plus the etch and stack dep that we've always had, as well as nanolization.

So, it's one of growth in NAND demand, but also growth in Lam's portfolio and served market as well, opportunities ahead. So, I think that that bodes well for us once the NAND business itself starts to recover.

Q - Joseph Quatrochi {BIO 18961101 <GO>}

Thanks for that. That's a helpful color. And just as a quick follow-up, you talked about global mature node spending being roughly flat this year. Can you just help us understand just kind of how does that break down? I mean, I think clearly the non-China piece is pretty weak, but just any kind of color that you can help us parse that out.

A - Doug Bettinger {BIO 16159085 <GO>}

Yeah, you kind of answered your own question, Joe. China's decent right now. Outside of China, it's pretty soft. And I think you understand what's going on. There's inventory that's built up, four days inventories still need to come down, and then mature node, analog, industrial, automotive space, and investment won't meaningfully occur until that gets adjusted. So, that's kind of what's going on. You sort of answered your own question.

Q - Joseph Quatrochi {BIO 18961101 <GO>}

Fair enough. Thanks.

A - Doug Bettinger {BIO 16159085 <GO>}

Thanks, Joe.

Operator

The next question is a follow-up question from Krish Sankar. Please go ahead.

Q - Krish Sankar {BIO 16151788 <GO>}

Yeah, hi. Thanks for taking my follow-up. Doug, I just have a quick follow-up. It's a question on inventory. You spoke about bringing on inventory turns down. I'm kind of curious, like how to think about inventory next year, especially if you're planning for a strong WFE year in 2025, or do you think most of this inventory would be in a bin? How to think about inventory next year? Thank you.

A - Doug Bettinger {BIO 16159085 <GO>}

Yes, listen, if you think back to when business declined for us last year, a big part of what fell off pretty rapidly was our NAND business.

And so, a lot of the inventory that we still have sitting on the balance sheet will support NAND. So, assuming NAND is stronger next year, and it will be, we will consume that NAND inventory that's been sitting on the balance sheet for a while. What will offset that to a certain extent is growth elsewhere, where we'll need to procure new inventory.

So, I'm not ready to give you an inventory forecast quite yet, but we're continuing to work as we go through the remainder of this year to kind of bring it down. And then, what we do with next year will depend largely on the timing of business, the mix of business, the geographic distribution of business. But we'd like to get turns back to where they historically have been, and they're not there yet. So, that's how you should be thinking about it, Krish.

Q - Krish Sankar {BIO 16151788 <GO>}

Thank you very much. Very helpful. Thanks, Doug.

A - Doug Bettinger {BIO 16159085 <GO>}

Thanks.

Operator

The next question is from Melissa Weathers with Deutsche Bank. Please go ahead.

Q - Melissa Weathers {BIO 21061067 <GO>}

Hi there. Thank you for taking my question. I wanted to ask on the leading-edge, and specifically gate-all-around nodes. I don't think I heard an update to your \$1 billion gate-all-around revenues in 2024 target. So, is that still the case? And then, as we think about those nodes ramping through next year, like, what's the trajectory of that ramp that we should be thinking about as you move from pilot lines into high-volume manufacturing?

A - Tim Archer {BIO 7502648 <GO>}

Yeah, thanks, Melissa. I mean, we always make choices about what we do and don't include in the prepared remarks, and the gate-all-around fell out this time, just no intended message.

In fact, if I look at what happened in the quarter Lam, again, as we've talked on previous calls, is really positioned quite nicely with our forward-looking etch and dep portfolios. We really targeted those kind of markets with new tools in Selective etch, which is a market we hadn't been in before. And in fact, in the quarter, we had additional Selective etch wins for gate-all-around at multiple customers.

We targeted investment in ALD films that are specifically needed for things like spacers and gate-all-around, and we had additional wins in the quarter for those films. Often with gate-all-around, we think of those technologies -- they're obviously different technologies, but kind of occurring at the same node. The backside power delivery, that's an area that is really right in our sweet spot in terms of deposition and etch. And in the quarter, we had wins in backside power in ALD oxide.

And so I would say, net-net, I mean, given those wins and what you're hearing from the end markets about the need for high-power computing demand for AI, I think the \$1 billion forecast we gave you, certainly as we move through next year, would be going higher.

And so, again, that's a combination of rising demand, but also a product portfolio that's both expanding and one in which we're winning share. And so, there was no message about leaving it off, but thank you for asking the question so we could get that in right at the end.

Q - Melissa Weathers {BIO 21061067 <GO>}

Perfect. Yeah, thank you. I'll stop my questions there.

A - Doug Bettinger {BIO 16159085 <GO>}

Okay.

A - Tim Archer {BIO 7502648 <GO>}

Great. Thank you.

A - Doug Bettinger {BIO 16159085 <GO>}

Awesome. Operator, I think that concludes our call. I want to thank everybody for joining us today. We'll look forward to seeing you at a variety of conferences and interactions as we go through the remainder of the quarter. Appreciate it.

A - Tim Archer {BIO 7502648 <GO>}

Thank you.

Operator

The conference is now concluded. Thank you for attending today's presentation. You may now disconnect.

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